

# Successful resolution of dengue-associated retinopathy following intravitreal Anti-VEGF therapy

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## ABSTRACT

Dengue fever (DF) is primarily recognized for its systemic manifestations; however, ocular involvement remains uncommon and is often overlooked. Dengue-associated retinopathy, a rare manifestation with poorly understood pathophysiological mechanisms, presents both diagnostic and therapeutic challenges. Although most patients recover spontaneously, some experience persistent visual impairment. We report a case of dengue-associated retinopathy with ischemic macular edema that progressed over 2 months following a diagnosis of DF. Owing to its ischemic nature and the patient's relative contraindications to intravitreal and systemic corticosteroids, intravitreal anti-vascular endothelial growth factor (anti-VEGF) therapy was administered, leading to the resolution of macular edema and improvement in visual function. To our knowledge, this is the first reported case demonstrating successful resolution of dengue-associated retinopathy with intravitreal anti-VEGF therapy. These findings suggest that anti-VEGF therapy may serve as an alternative treatment option for dengue-related ocular complications.

## 1. Introduction

Dengue fever (DF), the fastest-spreading mosquito-borne viral disease, has nearly doubled in global incidence between 1990 and 2021, with the highest burden concentrated in tropical and subtropical regions [1]. Ocular complications of dengue include subconjunctival hemorrhage, retinopathy, maculopathy, and optic neuropathy, with reported incidence rates ranging widely from 7 % to 87 % [2,3]. Dengue retinopathy results from altered expression of adhesion molecules in retinal endothelial cells and the loss of retinal pigment epithelium integrity, providing a molecular basis for dengue virus (DENV)-associated retinal vasculopathy and macular edema, respectively [4]. Progression to severe dengue is associated with the overproduction of inflammatory mediators, including cytokines, chemokines, and growth factors. Vascular endothelial growth factor (VEGF) has been identified as a biomarker of disease severity that increases endothelial permeability and gap formation, thereby mimicking the vascular leakage and inflammation observed in dengue [5]. Corticosteroids are currently the

mainstay of treatment for dengue-related ocular complications, although several studies have also shown that intravenous immunoglobulin therapy is effective [2]. Although anti-VEGF therapy shows promise in reducing vascular leakage and inflammation in dengue retinopathy, only one documented case, co-infected with DF and malaria, has been published to date, and it reported poor visual outcomes [6]. Here, we report the first case of successful treatment of dengue-associated retinopathy with intravitreal anti-VEGF therapy.

## 2. Case report

A 67-year-old man with a history of alcoholic liver cirrhosis post-liver transplantation, diabetes mellitus, hypertension, hyperlipidemia, and chronic kidney disease presented to the emergency department with a two-day history of fever. Although he had traveled from Central Taiwan, a region with low dengue endemicity, his presentation coincided with the 2023 dengue outbreak in southern Taiwan, particularly in Tainan City and Kaohsiung City, which together accounted for 93.3 %

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of all reported cases that year [7]. Dengue testing was performed in accordance with the protocol for febrile patients during the outbreak period. DF due to DENV-1 infection was confirmed by positive results for the dengue non-structural protein 1 (NS1) antigen and serotype-specific real-time polymerase chain reaction (RT-PCR) in serum. Initial laboratory tests revealed marked leukopenia (2400/ $\mu$ L), thrombocytopenia (101,000/ $\mu$ L), and mildly elevated liver enzyme levels. Serial complete blood counts during hospitalization showed a progressive decline in platelet count, reaching a nadir on the fourth day (32,000/ $\mu$ L) before gradually returning to normal. The patient was discharged after a 7-day course of treatment.

However, shortly after discharge, the patient experienced a sudden onset of bilateral blurred vision, which persisted for 2 months before he presented to our ophthalmology clinic. Ocular examination revealed a best-corrected visual acuity (BCVA) of 20/100 in both eyes, with no relative afferent pupillary defect. Fundus examination of both eyes showed mild tortuosity and dilatation of the retinal veins, peripapillary blot- and flame-shaped hemorrhages, and peripapillary cotton-wool spots (Fig. 1a). Notably, despite his diabetic status, fundus photographs taken 8 months earlier showed unremarkable findings (Fig. 1b). Fluorescein angiography (FA) demonstrated early peripapillary dot hyperfluorescent lesions surrounded by blocked hypofluorescent areas (Fig. 1c). In the late phase, diffuse perivascular leakage and patchy areas of capillary nonperfusion near the fovea were observed (Fig. 1d). Moreover, the absence of microaneurysms on FA indicated a low likelihood of diabetic retinopathy. Optical coherence tomography (OCT) confirmed the presence of macular edema in both eyes (Fig. 2a).

Further assessment was conducted to identify potential risk factors for venous stasis retinopathy. His systemic diseases were well controlled, and laboratory results showed normal levels of anti-cardiolipin antibodies, protein C, protein S, and antithrombin activity. Therefore, bilateral dengue-associated retinopathy was diagnosed, and treatment strategies were discussed with the patient. Given his phakic status and borderline optic disc cupping, the patient declined intravitreal corticosteroid therapy due to concerns about cataract progression and the risk of developing glaucoma. He also declined systemic corticosteroid therapy because he was already receiving multiple immunosuppressants due to his post liver transplantation status and was at increased risk of hyperglycemia due to diabetes. Considering the ischemic nature of his dengue retinopathy, an intravitreal injection (IVI) of bevacizumab (1.25 mg) was administered to both eyes. One month

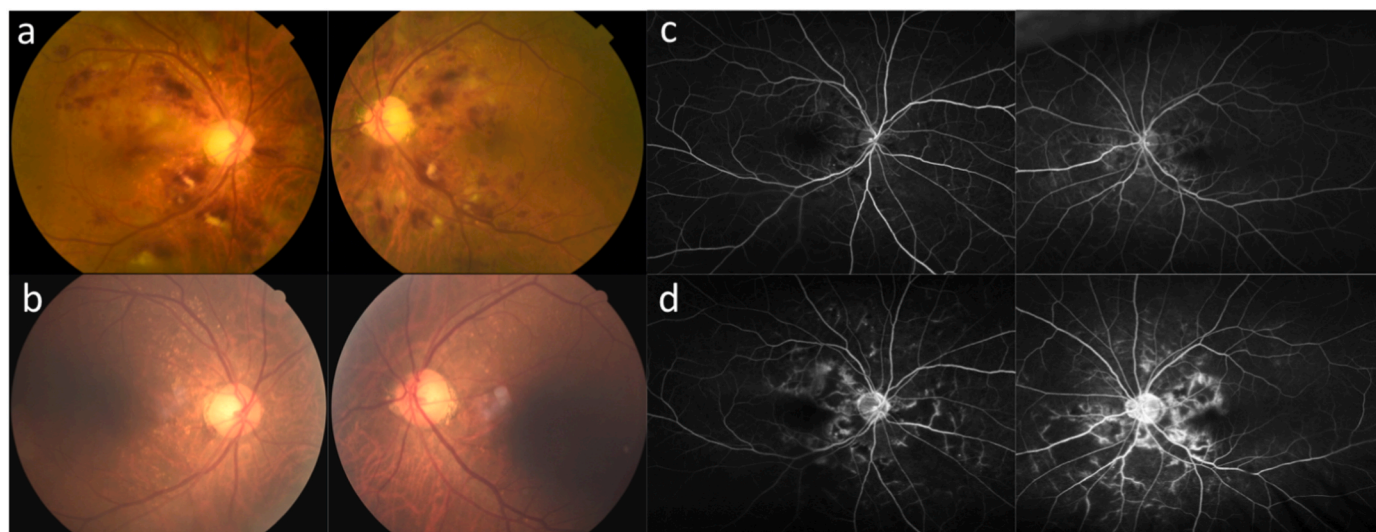
after IVI treatment, his BCVA improved to 20/70 in both eyes. The fundus showed improved venous contour and decreased retinal hemorrhage, and OCT demonstrated resolution of bilateral macular edema (Fig. 2b). His macular edema recurred 5 months after the first anti-VEGF injection (Fig. 2c); therefore, a second round of bilateral bevacizumab injections was administered. The patient subsequently remained stable without macular edema (Fig. 2d) and achieved a BCVA of 20/30 in the right eye and 20/70 in the left eye at his most recent follow-up, 1 year after the last intravitreal anti-VEGF injection.

### 3. Discussion

Dengue can lead to a range of ocular manifestations, with prevalence rates reported between 7 % and 40 %. Among these, dengue-associated retinopathy is relatively uncommon, occurring in approximately 10 % of hospitalized patients with DF. Some patients with dengue maculopathy experience spontaneous visual recovery, while others show poor response to treatment [8]. Our patient presented with dengue-associated retinopathy and ischemic macular edema and, to our knowledge, represents the first reported case successfully managed with intravitreal anti-VEGF injections.

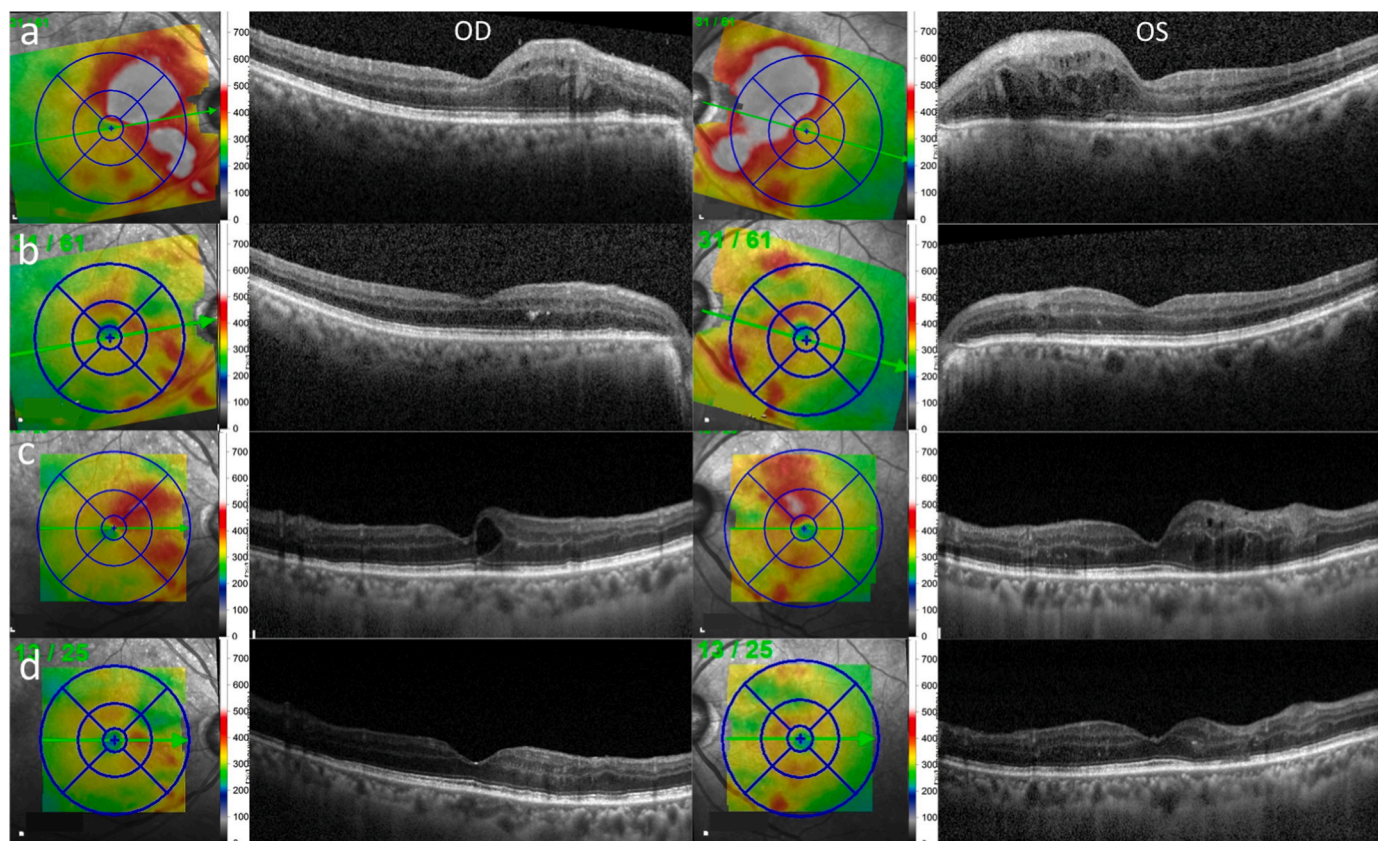
Dengue retinopathy typically presents bilaterally and asymmetrically. It is characterized by dot-blot or flame-shaped retinal hemorrhages, macular edema, cotton-wool spots, Roth spots, parafoveal yellow spots, microaneurysms, and arteriolar sheathing [9]. The most frequent posterior segment findings include macular edema, macular hemorrhage, and foveolitis [3]. Our patient exhibited bilateral venous stasis retinopathy-like features, including retinal venous tortuosity and dilatation, peripapillary retinal hemorrhages, cotton-wool spots, and macular edema. Velaitham et al. reported a similar case of central retinal vein occlusion concomitant with DF that eventually progressed to proliferative retinopathy [10].

Although the pathophysiology of dengue retinopathy remains unclear, it is generally considered to be inflammatory or occlusive in nature. This theory is supported by observations that visual symptoms often develop after systemic manifestations, typically within one day of the platelet nadir during dengue infection. It has been proposed that elevated interleukin (IL)-6 levels or molecular mimicry involving dengue viral structural proteins may trigger autoantibody formation against endothelial cells and platelets [11]. Another proposed mechanism is that anti-NS1 antibody cross-reactivity with vascular



**Fig. 1.** Clinical presentations of the patient. Fundus photos of both the patient's eyes initially demonstrated (a) peripapillary blot- and flame-shaped hemorrhages with cotton-wool spots, which was normal during (b) fundus photos taken eight months ago. Fundus fluorescein angiography of both eyes shows (c) peripapillary dot hyperfluorescent lesions surrounded by blocked hypofluorescent areas in early phase, and (d) peripapillary diffuse perivascular leakage along with patchy areas of capillary nonperfusion noted near the fovea in late phase.





**Fig. 2.** Spectral-domain optical coherence tomography (SD-OCT) of both eyes revealed (a) macular edema and increased reflectivity signal in the inner retina during initial presentation, (b) which resolved after first round of intravitreal Bevacizumab injections bilaterally. Five months later, (c) macular edema recurred, thus a second round of intravitreal Bevacizumab was arranged, and his macular status remained stable at (d) one-year follow-up.

endothelium leads to endothelial injury, reducing thrombomodulin expression and promoting thrombosis [12]. In our patient, blurred vision developed 3 days after the lowest platelet count, which may be attributed to delayed recognition of symptoms until they became more progressive and bilateral.

There is currently no established cure for dengue maculopathy, and management is primarily supportive. Corticosteroids have been used in some cases, although their efficacy remains unproven [13,14]. While the overall prognosis of ocular complications is generally favorable and may resolve spontaneously, approximately 5–8 % of affected eyes can develop permanent visual loss. Even with aggressive systemic corticosteroid therapy, patients presenting with severe dengue retinopathy and poor initial vision often show limited improvement [8,9]. Our patient declined both intravitreal and systemic corticosteroid treatment due to concerns regarding cataract formation and glaucoma progression, as well as consideration of his underlying medical conditions. Therefore, given the ischemic nature of his dengue retinopathy, we opted for an alternative treatment approach with intravitreal anti-VEGF therapy.

VEGF has been implicated as a key mediator in the immunopathogenesis of severe dengue. Targeting VEGF through anti-VEGF therapy may help preserve endothelial integrity and limit plasma leakage by inhibiting VEGF activity [5,15]. VEGF also plays a role in inflammatory responses, potentially through interactions with cytokines such as tumor necrosis factor (TNF)- $\alpha$ , IL-1 $\beta$ , and IL-6. Consequently, anti-VEGF therapy may attenuate inflammation by downregulating pro-inflammatory cytokine production, leading to reduced endothelial activation and damage [5,16]. Bevacizumab, a recombinant, humanized monoclonal antibody that binds to and neutralizes VEGF, is widely recognized as an effective treatment for choroidal neovascularization and macular edema [17]. A prior report by Rani et al. described a case of dengue retinopathy treated with bevacizumab; however, that patient had coexisting malaria,

a poor visual outcome, and limited clinical documentation [6]. In our patient, intravitreal bevacizumab treatment led to the resolution of macular edema and a marked improvement in visual acuity. Although macular edema recurred 5 months after the initial injection, his vision and macula condition remained stable without additional treatment during 1 year of follow-up.

To the best of our knowledge, this represents the first reported case of dengue retinopathy successfully treated with intravitreal anti-VEGF therapy. Further studies are warranted to clarify its therapeutic role; however, this approach may provide a localized, timely, and effective treatment option for patients with dengue retinopathy.

#### CRedit authorship contribution statement

**Yu-Chieh Chang:** Writing – original draft, Investigation, Conceptualization. **Yu-Ting Hsiao:** Writing – original draft, Visualization, Investigation, Conceptualization. **Hsi-Kung Kuo:** Writing – review & editing, Investigation, Conceptualization. **Hsiu-Mei Huang:** Writing – review & editing, Investigation, Conceptualization. **Jong-Jer Lee:** Writing – review & editing, Visualization, Supervision, Conceptualization.

#### Ethics approval and consent to participate

The study was approved by the ethics committee of medical ethics and human experiments of Chang Gung Memorial Hospital (IRB approval no. 202500959B0) and complied with the tenets of the Declaration of Helsinki. The patient was informed and signed a written informed consent.

## Data availability statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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